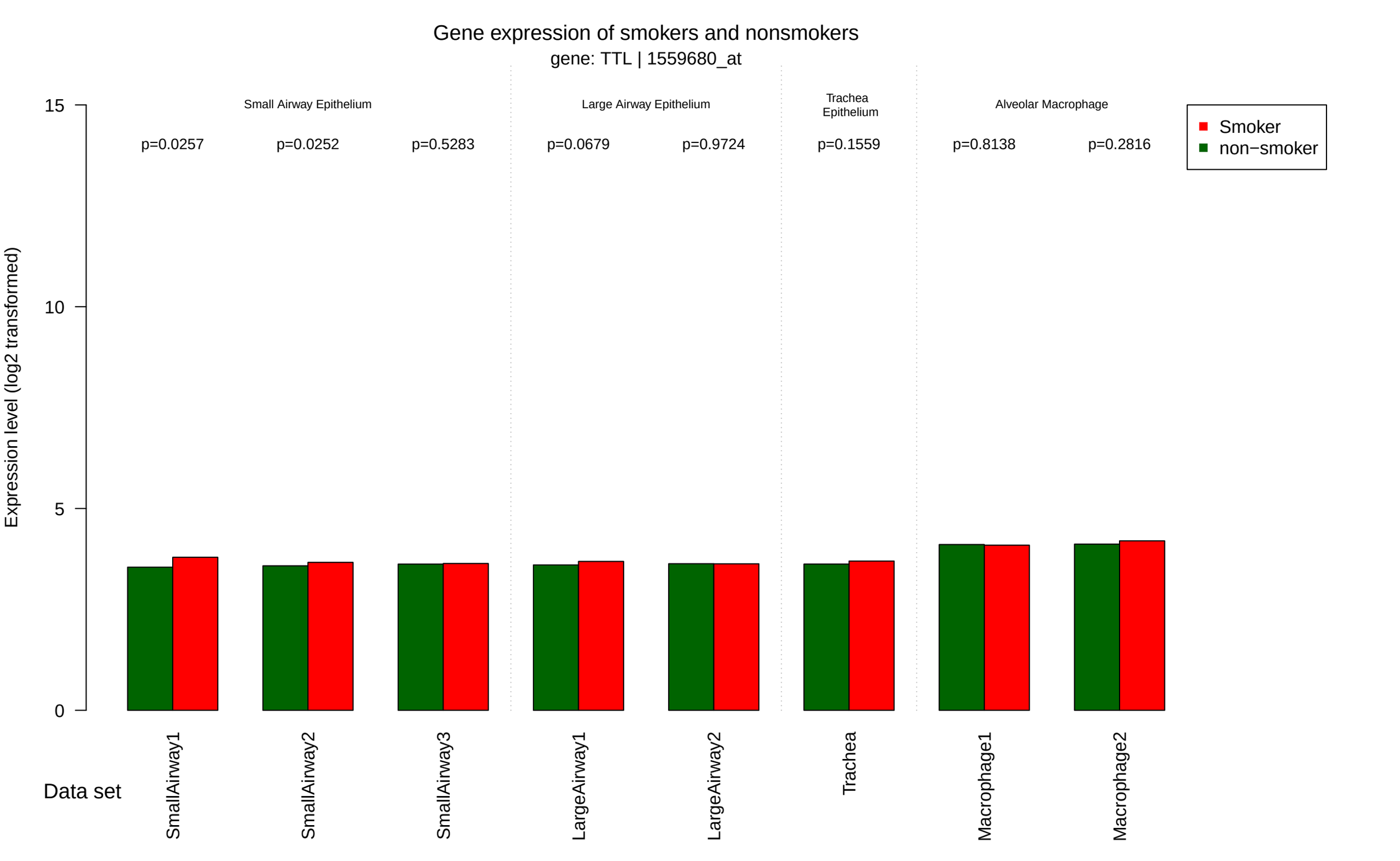
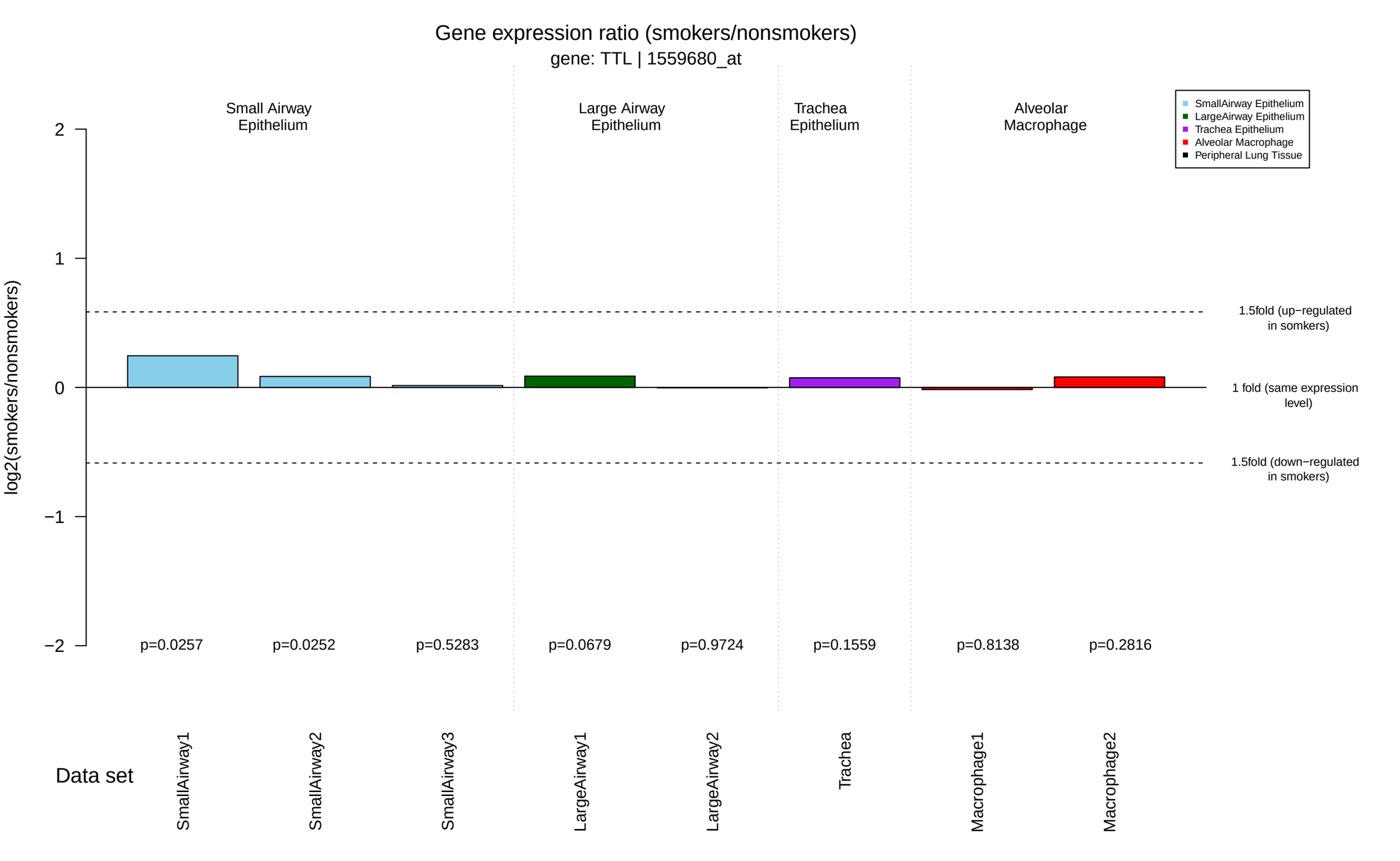
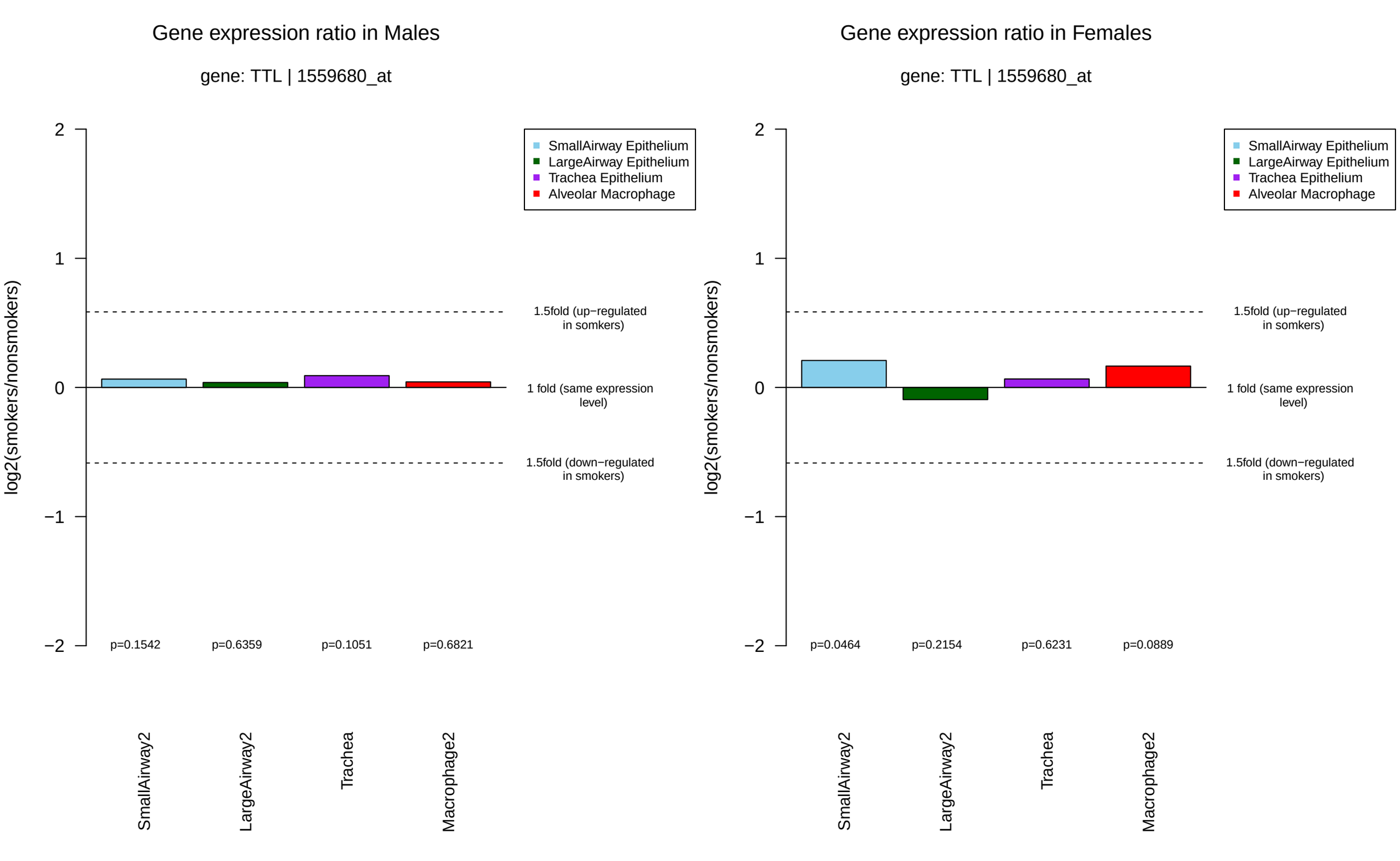
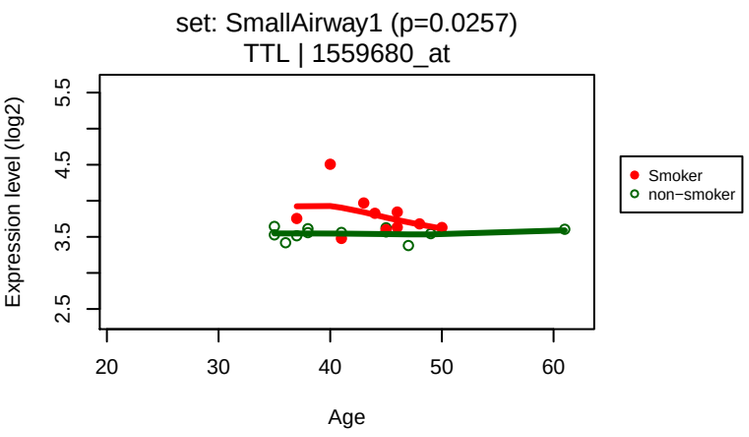
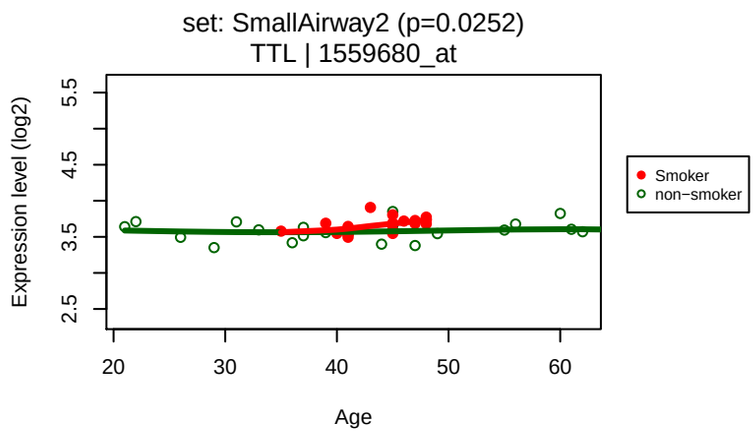
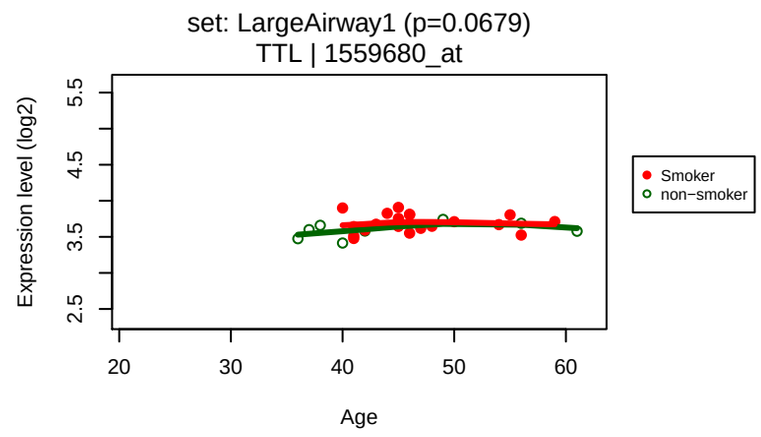

Gene expression of smokers and nonsmokers
gene: TTL | 1559680_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0257
p=0.0252
p=0.5283
p=0.0679
p=0.9724
p=0.1559
p=0.8138
p=0.2816
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

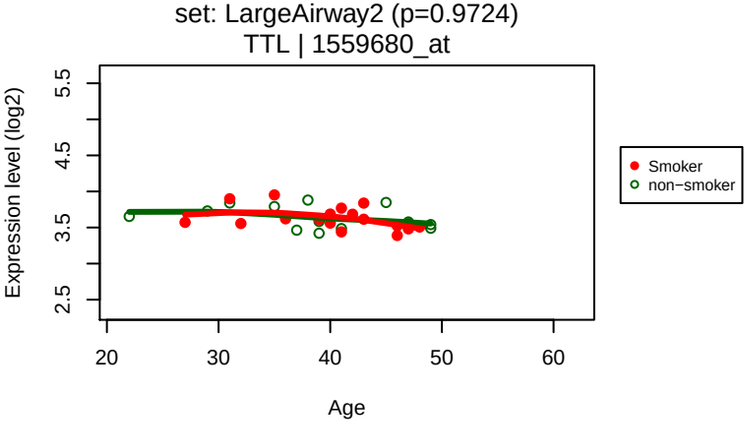

Gene expression ratio (smokers/nonsmokers)
gene: TTL | 1559680_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=0.0252
p=0.5283
p=0.0679
p=0.9724
p=0.1559
p=0.8138
p=0.2816
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

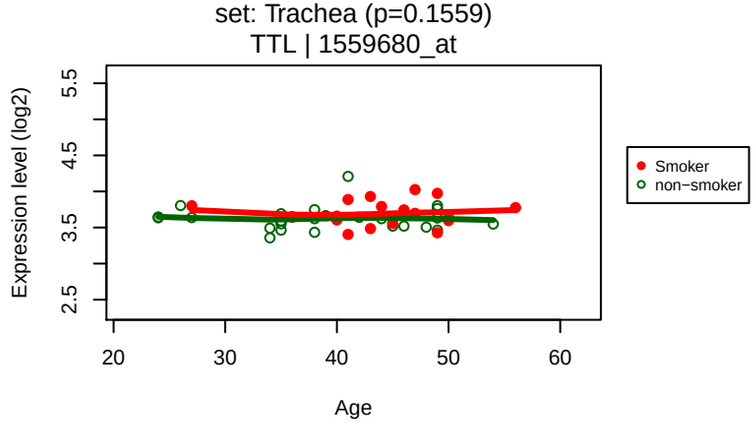

Gene expression ratio in Males
gene: TTL | 1559680_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1542
p=0.6359
p=0.1051
p=0.6821
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TTL | 1559680_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0464
p=0.2154
p=0.6231
p=0.0889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

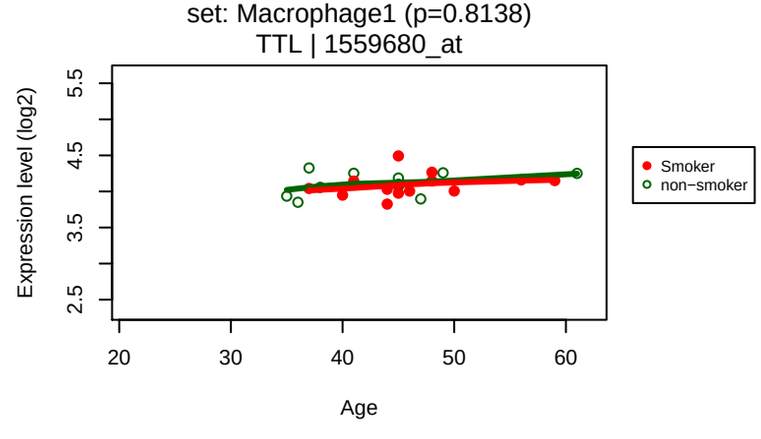

set: SmallAirway1 (p=0.0257)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker

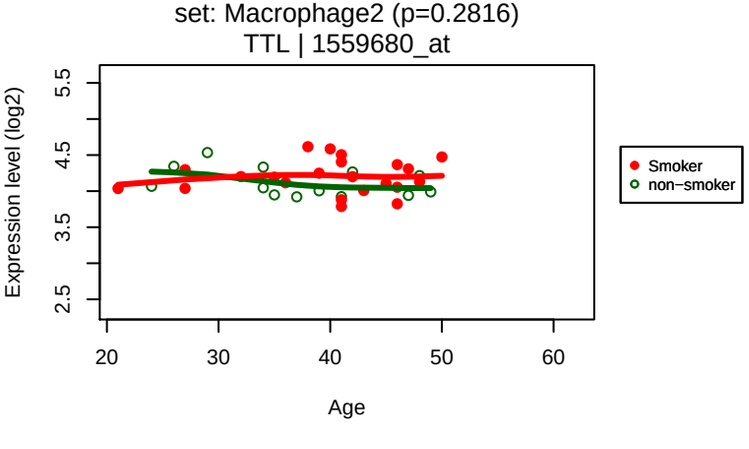

set: SmallAirway2 (p=0.0252)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker

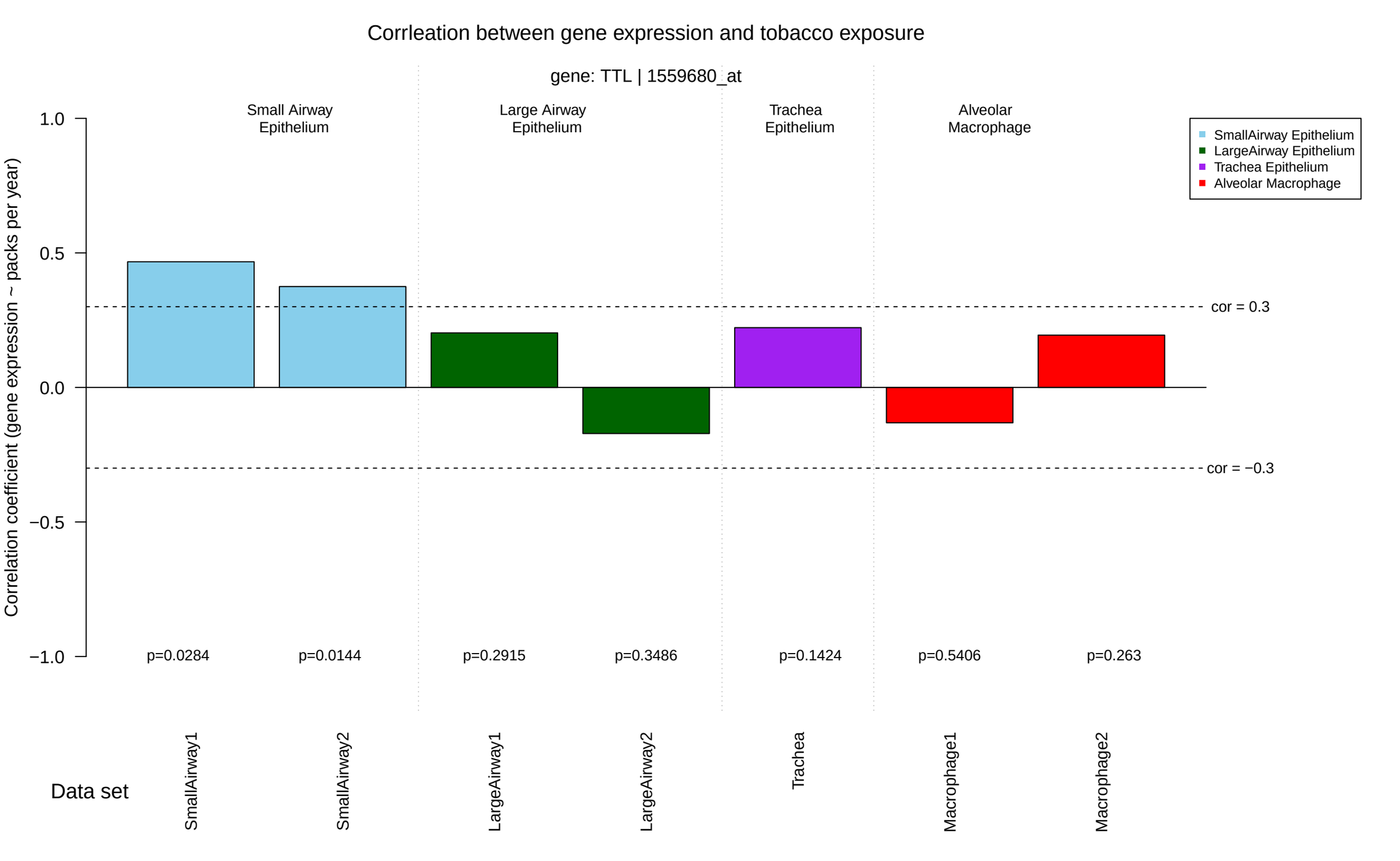

set: LargeAirway1 (p=0.0679)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9724)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1559)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8138)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2816)
TTL | 1559680_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TTL | 1559680_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0284
p=0.0144
p=0.2915
p=0.3486
p=0.1424
p=0.5406
p=0.263
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2